

International Conference on Advancement in Cyber Security and Digital Forensics



Special Session: Secured Sensor Fusion for Intelligent EVs

Overview

Modern electric vehicles are rapidly evolving into sensor-rich intelligent systems, equipped with LiDAR, radar, cameras, ultrasonic sensors, IMUs, GPS, and V2X communication modules. However, individual sensors are inherently noisy, uncertain, and limited in perception scope. Sensor fusion provides the critical layer of intelligence that integrates heterogeneous data streams into a coherent, reliable, and context-aware understanding of the environment.

Topics of Interest

- Multi-sensor data fusion in electric and autonomous vehicles
- LiDAR–Radar–Camera fusion techniques
- Sensor fusion for ADAS and autonomous navigation
- Kalman filtering, particle filtering, and probabilistic fusion
- Deep learning-based sensor fusion models
- Transformer and graph-based fusion architectures
- Sensor fusion under uncertainty and noisy environments
- Edge AI for real-time fusion in EV systems
- Cybersecurity in sensor fusion (spoofing, adversarial attacks)
- V2X-based cooperative perception and fusion
- Fusion-enabled digital twins for EVs
- Sensor fusion for battery monitoring and thermal systems
- Explainable AI in sensor fusion systems

Session Organizers



Dr. Souvik Ganguli
Thapar Institute of Engineering and
Technology, Patiala, Punjab
Email: souvik.ganguli@thapar.edu



Dr. Maitrayee Chakrabarty
Narula Institute of Technology, Agarpara,
West Bengal
Email: moitry29@gmail.com